

VIRTUAL TRY ON

AI/ML INTERN

1. Introduction to 2D Virtual Try-On

2D Virtual Try-On is an Artificial Intelligence and Computer Vision technology that allows users to digitally wear clothes on their images without physically trying them. This technology is mainly used in fashion e-commerce applications to improve customer experience and reduce product returns. The system works by detecting body landmarks such as shoulders and torso positions using AI models and then overlays clothing images on the detected body area. In this project, OpenCV and MediaPipe were used to create a basic virtual try-on prototype that aligns garments according to human body posture.

2. What We Did to Get This Output

- The following steps were performed to generate the virtual try-on output:
- Step 1:** Uploaded the person image and cloth image into Google Colab.
 - Step 2:** Used OpenCV to preprocess images and resize them for proper alignment.
 - Step 3:** Used MediaPipe Pose Detection to detect body landmarks such as shoulders and hips.
 - Step 4:** Calculated shoulder width and torso size dynamically for proper garment fitting.
 - Step 5:** Resized the clothing image according to body measurements.
 - Step 6:** Removed cloth background using masking techniques.
 - Step 7:** Overlayed the cloth image on the user image using OpenCV blending operations.
 - Step 8:** Generated the final virtual try-on output image.

3. Technologies Used

Technology	Purpose
OpenCV	Image Processing and Overlay
MediaPipe	Pose Detection and Landmark Extraction
NumPy	Array and Image Operations
Google Colab	Execution Environment

4. Python Code

```
# FINAL 2D VIRTUAL TRY-ON CODE
import cv2
```

```

import mediapipe as mp
import numpy as np
from google.colab import files
import matplotlib.pyplot as plt

uploaded = files.upload()

file_names = list(uploaded.keys())

person_path = None
cloth_path = None

for file in file_names:
    lower = file.lower()
    if "person" in lower:
        person_path = file
    elif "cloth" in lower or "shirt" in lower:
        cloth_path = file

person_img = cv2.imread(person_path)
cloth_img = cv2.imread(cloth_path)

person_img = cv2.resize(person_img, (500, 700))

rgb = cv2.cvtColor(person_img, cv2.COLOR_BGR2RGB)

mp_pose = mp.solutions.pose

with mp_pose.Pose(
    static_image_mode=True,
    model_complexity=1,
    min_detection_confidence=0.5
) as pose:
    results = pose.process(rgb)

landmarks = results.pose_landmarks.landmark

left_shoulder = landmarks[11]
right_shoulder = landmarks[12]

left_hip = landmarks[23]
right_hip = landmarks[24]

h, w, _ = person_img.shape

left_x = int(left_shoulder.x * w)
right_x = int(right_shoulder.x * w)

left_y = int(left_shoulder.y * h)

hip_y = int(((left_hip.y + right_hip.y) / 2) * h)

shoulder_width = abs(right_x - left_x)

cloth_width = int(shoulder_width * 1.8)

torso_height = abs(hip_y - left_y)

cloth_height = int(torso_height * 1.2)

cloth_resized = cv2.resize(cloth_img, (cloth_width, cloth_height))

x_offset = max(0, left_x - int(cloth_width * 0.25))
y_offset = max(0, left_y - 20)

cloth_height = min(cloth_height, person_img.shape[0] - y_offset)
cloth_width = min(cloth_width, person_img.shape[1] - x_offset)

cloth_resized = cv2.resize(cloth_resized, (cloth_width, cloth_height))

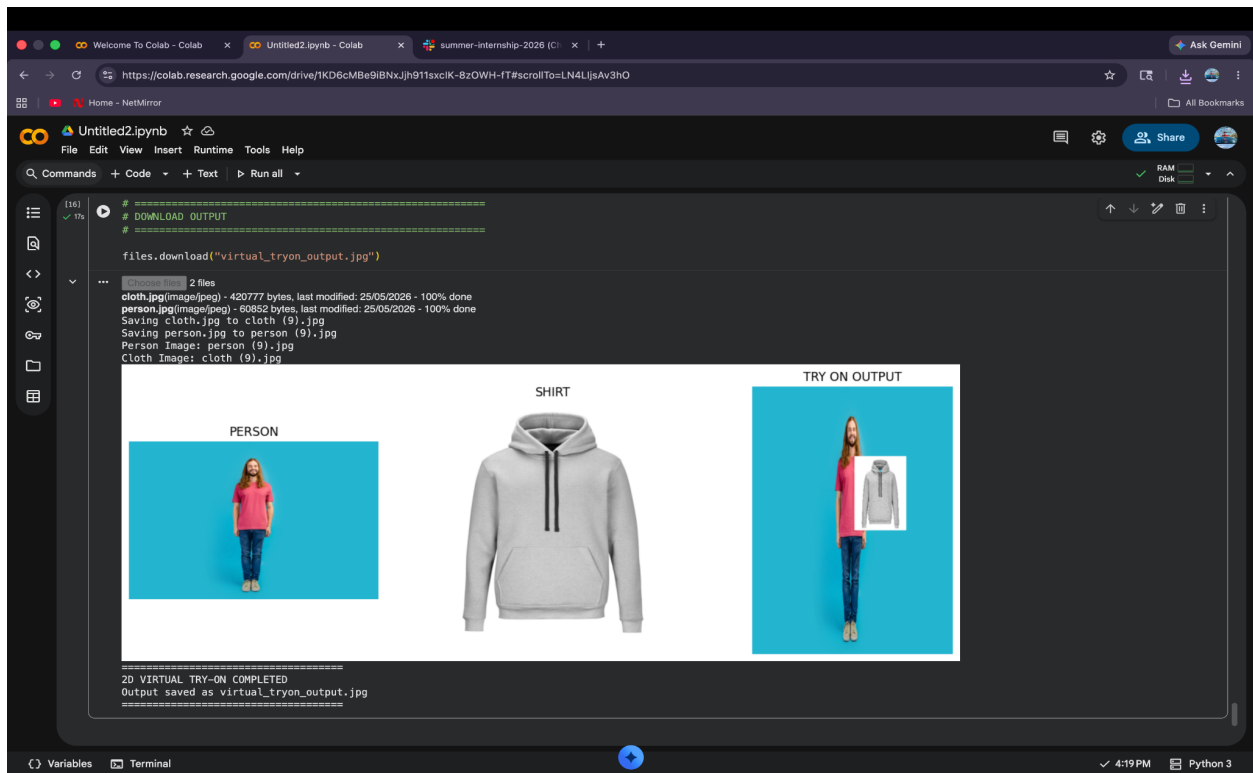
roi = person_img[
    y_offset:y_offset + cloth_height,
    x_offset:x_offset + cloth_width
]
gray = cv2.cvtColor(cloth_resized, cv2.COLOR_BGR2GRAY)

```

```
_, mask = cv2.threshold(gray, 20, 255, cv2.THRESH_BINARY)
mask_inv = cv2.bitwise_not(mask)
person_bg = cv2.bitwise_and(roi, roi, mask=mask_inv)
shirt_fg = cv2.bitwise_and(cloth_resized, cloth_resized, mask=mask)
combined = cv2.add(person_bg, shirt_fg)
person_img[
    y_offset:y_offset + cloth_height,
    x_offset:x_offset + cloth_width
] = combined
cv2.imwrite("virtual_tryon_output.jpg", person_img)
print("2D Virtual Try-On Completed")
```

5. Output Screenshots





Conclusion

This project successfully demonstrates a basic AI-powered 2D Virtual Try-On system using OpenCV and MediaPipe. The model detects body landmarks and overlays garments according to the user's body posture. Although the current implementation uses simple overlay techniques, advanced deep learning models such as HR-VITON can further improve realism and cloth fitting quality.